
Bayesian Inference

A Complete Course Notebook

From Classical Inference to Conjugate Priors, Exchangeability,
de Finetti's Theorem, and Objective Bayesian Analysis

Coverage

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| Week 1 | Classical vs. Bayesian Inference; Bayes' Theorem |
| Week 2 | Steps of Bayesian Analysis; Exchangeability; de Finetti; Beta-Bernoulli |
| Week 3 | Conjugate Priors: Beta-Bernoulli & Normal-Normal |
| Week 4 | Improper Priors; Jeffreys Prior; Kullback-Leibler Divergence |

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Chapter 1

Week 1: Classical Inference, Bayes' Theorem, and the Bayesian Paradigm

1.1 Classical (Frequentist) Inference: An Overview

Classical statistical inference treats the unknown quantity of interest, θ , as a *fixed but unknown constant*, and uses the randomness of the data $X_1, \dots, X_n$ to quantify uncertainty. In contrast, **Bayesian inference** treats θ itself as a random quantity possessing a probability distribution, which is updated in light of data.

Classical inference: estimation

Given $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f(\cdot | \theta)$, classical **estimation** consists of

- a **point estimator**, e.g. the sample mean $\bar{X} = \frac{\sum_i X_i}{n}$ as an estimator of μ when $X_i \stackrel{\text{iid}}{\sim} N(\mu, 1)$;
- a **confidence interval**, e.g. the classical 95% CI for a normal mean,

$$\left(\bar{X} - 1.96 \frac{s}{\sqrt{n}}, \bar{X} + 1.96 \frac{s}{\sqrt{n}} \right).$$

A key idea behind evaluating any estimator $T(X_n)$ is to imagine repeating the experiment many times, $Y_1, \dots, Y_M$ being M independent replications, each yielding an estimate $T^{(1)}(X_n), \dots, T^{(M)}(X_n)$, with sample mean $\bar{T}_M = \frac{1}{M} \sum_{i=1}^M T^{(i)}(X_n)$. The quality of T is then judged via

$$E(T(X)), \quad \text{Var}(T(X)) = \frac{1}{M} \sum_{i=1}^M (T^{(i)}(X_n) - \bar{T}_M)^2,$$

which together determine the **Mean Squared Error** (MSE) of the estimator, the standard frequentist yardstick of estimator quality.

Interpretation

Classical inference is fundamentally about the sampling distribution of an estimator or statistic over hypothetical repetitions of the data-generating experiment — *not* about a probability statement on θ itself.

1.2 Hypothesis Testing: Practical vs. Statistical Significance

1.2.1 The Classical Test

Suppose $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\theta, 1)$ and we wish to test

$$H_0 : \theta = 0 \quad \text{vs.} \quad H_1 : \theta \neq 0, \quad \alpha = 0.05.$$

The classical test rejects H_0 if

$$\text{Reject } H_0 \text{ if } \sqrt{n} |\bar{x}| > 1.96.$$

1.2.2 A Cautionary Example

Take a true effect size $\theta = 10^{-10}$, and a huge sample size $n = 10^{24}$. Then

$$\sqrt{n} |\bar{x}| \approx 10^{12} \cdot 10^{-10} \gg 1.96,$$

so H_0 is rejected with overwhelming statistical significance — even though θ is so close to 0 that it is of *no practical importance whatsoever*. This is the classic tension between **statistical significance** and **practical significance**: with enough data, a classical test will detect and reject even a scientifically meaningless departure from H_0 .

Bayesian contrast

The Bayesian framework naturally distinguishes the two notions of significance by directly reporting posterior probabilities of scientifically meaningful regions, e.g.

$$\pi(\theta \in (-10^{-5}, 10^{-5}) \mid X) \quad \text{or} \quad \pi(p \in (0.49, 0.51) \mid X),$$

rather than a binary reject/do-not-reject verdict.

1.2.3 Variance and the Danger of Random Sample Sizes

Consider an experiment where a coin toss determines the sample size: if heads, $n = 2$; if tails, $n = 1000$. If $X_i \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$, the (unconditional) variance of $\bar{X}$ is

$$V(\bar{X}) = \frac{1}{2} \left[\frac{\sigma^2}{2} + \frac{\sigma^2}{1000} \right] \approx \frac{\sigma^2}{4},$$

which is badly misleading as a description of the precision of $\bar{X}$ in either individual scenario ($n = 2$ or $n = 1000$) — a further illustration of how classical variance calculations can average over irrelevant hypothetical repetitions.

1.2.4 A Paradox with the Uniform Location Model

Let $X \sim U(\theta - \frac{1}{2}, \theta + \frac{1}{2})$, and consider $\pi(\theta \mid \text{Data})$. With a single pair of observations $X_1 = 2, X_2 = 2$ (so $X_2 - X_1 = 0$), the data pin down θ 's location interval essentially exactly, yet classical confidence procedures based on this model can produce intervals that fail to reflect this certainty — a preview of the paradoxes discussed further in Section 1.7.

1.3 Bayes' Theorem: The Disease-Diagnostic Example

Setup

A disease affects 2% of a population. A diagnostic test has:

- If a person has the disease, the test is positive with probability 0.95.
- If a person does *not* have the disease, the test is (falsely) positive with probability 0.10.

Let D = "person has the disease", T^+ = "test result is positive". Find $P(D | T^+)$.

Solution

$$P(D) = 0.02, \quad P(D^c) = 0.98 \quad [\text{prior probabilities}]$$

$$P(T^+ | D) = 0.95, \quad P(T^+ | D^c) = 0.10 \quad [\text{conditional probabilities / likelihoods}]$$

$$P(T^+) = P(D)P(T^+ | D) + P(D^c)P(T^+ | D^c) = (0.02)(0.95) + (0.98)(0.10) = 0.117 \quad [\text{marginal probability}]$$

By Bayes' theorem,

$$P(D | T^+) = \frac{P(D \cap T^+)}{P(T^+)} = \frac{P(T^+ | D)P(D)}{P(T^+)} = \frac{(0.95)(0.02)}{0.117} = 0.1624.$$

Why the result feels surprising

Even with a fairly accurate test (95% sensitivity), a positive result only raises the probability of disease from a 2% prior to a 16.24% posterior, because the disease is rare and false positives (10% of the large healthy population) numerically swamp the true positives from the small diseased population. This is the base-rate fallacy, and it is precisely what Bayes' theorem is designed to correct for.

1.4 The Three Prisoners Problem

Setup

Three prisoners, A , B , C , are told that two of them will be executed and one pardoned, each equally likely a priori: $P(A) = P(B) = P(C) = \frac{1}{3}$. Prisoner A asks the guard "will prisoner B be executed?" (call this event b), knowing the guard will truthfully name one of B or C who is *not* pardoned. If A is pardoned, the guard picks between naming B or C with probability $\frac{1}{2}$ each; if B is pardoned, the guard can never say b ($P(b | B) = 0$); if C is pardoned, the guard must say b ($P(b | C) = 1$).

What is $P(\text{pardon for } A | b)$? (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{2}{3}$ (d) 1

Solution

$$P(A, b) = P(A) P(b | A) = \frac{1}{3} \times \frac{1}{2}$$

$$P(b) = P(A)P(b | A) + P(B)P(b | B) + P(C)P(b | C) = \frac{1}{3} \times \frac{1}{2} + \frac{1}{3} \times 0 + \frac{1}{3} \times 1 = \frac{1}{3} \times \frac{3}{2} = \frac{1}{2}$$

$$P(A | b) = \frac{P(A, b)}{P(b)} = \frac{\frac{1}{3} \times \frac{1}{2}}{\frac{1}{2}} = \frac{1}{3}.$$

The answer is (a) $\frac{1}{3}$: learning that B will be executed gives *no* information about A 's own fate — it only resolves the ambiguity between B and C (whose posterior pardon probability rises to $\frac{2}{3}$). This is a classic illustration of updating correctly via conditioning rather than by naive intuition.

1.5 Course Structure and Reference Material

Grading: 25% Mid-semester exam • 50% End-semester exam • 25% Project / assignment / quizzes (an assignment on reading and applying a Bayesian application paper is part of this component).

Reference text: *Bayesian Data Analysis* (Gelman et al.). **Computational tools:** Stan (via the `rstan` and `pystan` interfaces).

1.6 Prior, Likelihood, and Posterior — Revisited

Returning to the disease-testing example, we now view the calculation through the prior/likelihood/posterior lens that will recur throughout the course:

$$\underbrace{P(D) = 0.02}_{\text{prior}}, \quad \underbrace{P(T^+ | D) = 0.95}_{\text{conditional / likelihood}}, \quad \underbrace{P(D | T^+) = 0.1624}_{\text{posterior}}.$$

Sensitivity to the prior

If instead the base rate were $P(D) = 0.20$ (so $P(D^c) = 0.80$), the same test characteristics give a posterior $P(D | T^+) = 0.704$ — a dramatically different answer, showing how strongly the posterior depends on the assumed prior probability of disease.

The general mechanics, applied to both hypotheses D and D^c , are:

$$P(D | T^+) = \frac{P(D) P(T^+ | D)}{P(T^+)}, \quad P(D^c | T^+) = \frac{P(D^c) P(T^+ | D^c)}{P(T^+)}.$$

Since these must sum to 1, it is often more convenient to work with the **unnormalized posterior** and normalize at the end. Writing $k = P(T^+)$,

$$P(D | T^+) \propto P(D) P(T^+ | D) = \frac{0.019}{k}, \quad P(D^c | T^+) \propto P(D^c) P(T^+ | D^c) = \frac{0.098}{k},$$

which is summarized in the single most important slogan of Bayesian statistics:

$$\text{Posterior} \propto \text{Prior} \times \text{Likelihood}$$

Connection to hypothesis testing

The same disease-testing structure can be recast as testing $H_0 : p = 0.95$ (test performs well under disease) vs. $H_1 : p = 0.10$ (test performs like a false-positive generator), with $P(T^+ | D) = 0.95$ and $P(T^+ | D^c) = 0.10$ playing the role of likelihoods under the two hypotheses — foreshadowing the posterior-odds framework of Week 2.

Prior sensitivity table

The table below shows how the posterior probability of H_0 (disease present) changes as the *subjective* prior probability changes, holding the likelihoods fixed:

$$P(H_0) = 0.02, \quad P(H_1) = 0.98 \Rightarrow P(H_0 | T^+) = 0.16$$

$$P(H_0) = 0.20, \quad P(H_1) = 0.80 \Rightarrow P(H_0 | T^+) = 0.70$$

$$P(H_0) = 0.50, \quad P(H_1) = 0.50 \Rightarrow P(H_0 | T^+) \approx 0.90$$

1.7 Paradoxes in the Classical Paradigm: The Uniform Location Example

Setup

Let $X \sim U(\theta - \frac{1}{2}, \theta + \frac{1}{2})$, and suppose we observe two iid draws X_1, X_2 . Define

$$y_1 = \min\{X_1, X_2\}, \quad y_2 = \max\{X_1, X_2\}.$$

A valid 50% confidence interval

For any true θ ,

$$P(y_1 < \theta < y_2 | \theta) = P(X_1 < \theta < X_2) + P(X_2 < \theta < X_1) = \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} = \frac{1}{2}.$$

Hence (y_1, y_2) is a valid 50% **confidence interval** for θ , in the classical, pre-data, repeated-sampling sense.

The paradox

Now suppose the *observed* data happen to be $x_1 = 7.4$, $x_2 = 8.0$, giving the interval $(7.4, 8.0)$. But because $\theta - \frac{1}{2} < x_1, x_2 < \theta + \frac{1}{2}$ for *every* valid θ under this model, we in fact know with **certainty** (probability 1, not 50%) that

$$\theta + \frac{1}{2} > 8.0 \Rightarrow \theta > 7.5, \quad \theta - \frac{1}{2} < 7.4 \Rightarrow \theta < 7.9,$$

so $\theta \in (7.5, 7.9)$ with absolute certainty, given this particular sample. Yet the procedure is labeled "50% confidence" because that is its long-run coverage rate *averaged over all possible data sets* — a coverage guarantee that says nothing about how confident we should be about θ *given the specific data actually observed*. This is precisely the gap that a posterior distribution $\pi(\theta | x_1, x_2)$ closes: it is a probability statement conditional on the data we actually saw, not on hypothetical repetitions of the experiment.

1.8 The Likelihood Principle: Binomial vs. Negative Binomial Sampling

Setup

Twelve independent coin tosses are performed and 9 heads are observed. We wish to test

$$H_0 : p = \frac{1}{2} \quad \text{vs.} \quad H_1 : p > \frac{1}{2}, \quad \alpha = 0.05.$$

Two different "experiments"

If $n = 12$ was fixed in advance (Binomial sampling):

$$L_1(p) = \binom{12}{9} p^9 (1-p)^3, \quad \text{p-value} = P(X \geq 9) = 0.075.$$

If sampling stopped as soon as the 9th head occurred (Negative Binomial sampling):

$$L_2(p) = \binom{11}{9} p^9 (1-p)^3, \quad \text{p-value} = P(X \geq 9) = 0.0325.$$

The paradox

The two experiments produced *numerically the same data* (9 heads, 3 tails) and *proportional* likelihood functions ($L_1(p) \propto L_2(p) \propto p^9(1-p)^3$), yet the classical p -values disagree (0.075 vs. 0.0325) — one leads to non-rejection at $\alpha = 0.05$, the other to rejection! The discrepancy arises entirely from the *stopping rule* (i.e., from outcomes that were never observed), which the classical p -value calculation must average over but which a likelihood-based/Bayesian analysis simply ignores.

The Likelihood Principle (LP)

In making inferences or decisions about θ after x is observed, all relevant information about θ contained in the experiment is contained in the **likelihood function** of the observed x — and only in its *shape* as a function of θ (i.e., proportional likelihoods carry identical information).

1.9 Advantages of the Bayesian Paradigm

- (1) Prior information can be formally incorporated into the analysis.
- (2) Inferences are conditional on the actual observed data, via $\frac{P(\theta)P(D_n | \theta)}{P(D_n)}$, rather than on hypothetical repetitions.
- (3) The paradigm automatically satisfies the Likelihood Principle.
- (4) A single posterior distribution answers estimation, testing, and prediction questions directly — there is no need for a separate machinery of hypothesis tests, confidence intervals, p -values, etc.
- (5) Bayesian inference does not rely on asymptotic (large-sample) approximations to be valid.

- (6) However complex the model, modern computational methods (MCMC, Stan, etc.) deliver arbitrarily accurate inference by computational means.

Chapter 2

Week 2: Steps of Bayesian Analysis, Exchangeability, and de Finetti's Theorem

2.1 The Three Steps of Bayesian Data Analysis

Overall goal

Bayesian data analysis is about finding the **conditional distribution of unknown (parameter) quantities, given the observed data**.

Step 1 — Setting up a full probability model. Construct a joint probability distribution for *all* observable and unobservable quantities, consistent with knowledge of the underlying scientific problem and the data-collection process.

Step 2 — Conditioning on observed data. The posterior distribution is the conditional distribution of the unobserved quantities given the observed data.

Step 3 — Evaluating the fit of the model.

Running example: coin tosses with unknown bias

For Step 1, take a sequence of coin tosses with unknown bias θ : what is *observable* is $y_1, \dots, y_n$; what is *unobservable* is θ ; and the joint distribution over everything is

$$p(\theta, y_1, \dots, y_n) = p(\theta) p(y_1, \dots, y_n \mid \theta).$$

For Step 3, the standard diagnostics (introduced later in the course) include **posterior predictive checks**, which compare data simulated from the fitted model against the actually observed data.

2.2 Posterior Odds vs. Prior Odds

Setup

Choose between two competing values/hypotheses θ_1 or θ_2 for a parameter (e.g. testing whether an estimator is unbiased, or comparing two models).

$$\text{Prior odds} = \frac{p(\theta_1)}{p(\theta_2)}, \quad \text{Posterior odds} = \frac{p(\theta_1 | y)}{p(\theta_2 | y)}.$$

Derivation

By Bayes' rule applied separately to each hypothesis,

$$\frac{p(\theta_1 | y)}{p(\theta_2 | y)} = \frac{\frac{p(\theta_1) p(y | \theta_1)}{p(y)}}{\frac{p(\theta_2) p(y | \theta_2)}{p(y)}} = \frac{p(\theta_1) p(y | \theta_1)}{p(\theta_2) p(y | \theta_2)} = \underbrace{\frac{p(\theta_1)}{p(\theta_2)}}_{\text{prior odds}} \times \underbrace{\frac{p(y | \theta_1)}{p(y | \theta_2)}}_{\text{likelihood ratio (Bayes factor)}}.$$

$$\frac{p(\theta_1 | y)}{p(\theta_2 | y)} \geq \frac{p(\theta_1)}{p(\theta_2)} \iff \frac{p(y | \theta_1)}{p(y | \theta_2)} \geq 1,$$

i.e. the posterior odds move away from the prior odds in the direction favored by the likelihood ratio: if the data are more likely under θ_1 than under θ_2 , the posterior odds shift toward θ_1 relative to the prior odds, and vice versa.

Asymptotic behaviour as $n \rightarrow \infty$

As the sample size n grows, the likelihood ratio dominates the fixed prior odds, and the posterior odds concentrate on whichever of θ_1, θ_2 is (closer to) the truth.

Exercise

State and prove this consistency result precisely: under repeated iid sampling, show that $p(y_1, \dots, y_n | \theta_1) / p(y_1, \dots, y_n | \theta_2) \rightarrow 0$ or ∞ almost surely, depending on which hypothesis actually generated the data (a Kullback–Leibler / law-of-large-numbers argument applied to the log-likelihood ratio).

2.3 Conditional Independence and Exchangeability

Motivating example

Consider observations $y_1, \dots, y_{10}$ from 10 people (e.g. heights in cm, in the range 100–200), with an 11th person's observation y_{11} to be predicted.

2.3.1 Conditional independence given θ

Given the parameter θ , a new observation is independent of the past observations:

$$p(y_{n+1} | \theta, y_1, \dots, y_n) = p(y_{n+1} | \theta).$$

Marginally (i.e. before conditioning on θ), however,

$$p(y_{n+1} | y_1, \dots, y_n) = f(y_1, \dots, y_n)$$

is *not* simply $p(y_{n+1})$: the past data inform beliefs about θ , which in turn inform the prediction of y_{n+1} . This is the basic mechanism of **learning from data** in the Bayesian framework.

2.3.2 Exchangeability

Definition 2.1 (Exchangeability). $y_1, \dots, y_n$ are **exchangeable** if their joint distribution $p(y_1, \dots, y_n)$ is invariant under any permutation π of $\{1, \dots, n\}$:

$$p(y_1, \dots, y_n) = p(y_{\pi(1)}, \dots, y_{\pi(n)}).$$

In particular, for a new observation the specific labeling/ordering of the conditioning variables does not matter:

$$p(y_{n+1} \mid y_1, \dots, y_n) = p(y_{n+1} \mid y_2, y_1, \dots, y_n).$$

Why this matters

Exchangeability is a weaker, more natural assumption to justify from modeling considerations directly than "iid" — yet (via de Finetti's representation theorem, below) it turns out to be *equivalent* to the existence of a parameter θ and a prior $p(\theta)$ such that, conditional on θ , the y_i are iid. This is the conceptual bridge between "exchangeable data" and "iid-given- θ data with a prior on θ ."

Exercise

Write out carefully why iid $\Rightarrow$ exchangeable but not conversely in general (an exchangeable sequence need not be iid unconditionally — only conditionally on θ , per de Finetti's theorem). A short two-observation counterexample (e.g. draws *without replacement* from a finite population) is a good self-check.

2.4 De Finetti's Representation Theorem

Setup

Consider binary sequences, e.g. $1, 0, 0, 1, 0, \dots$, with $\sum_{i=1}^n Y_i$ as the natural summary statistic.

Theorem 2.2 (de Finetti). *If $y_1, y_2, \dots$ is an (infinitely extendable) exchangeable sequence of binary outcomes, then there exists a distribution $p(\theta)$ such that*

$$p(y_1, \dots, y_n) = \int_0^1 \theta^{\sum_i y_i} (1 - \theta)^{n - \sum_i y_i} p(\theta) d\theta.$$

Equivalently: $Y_1, \dots, Y_n \mid \theta \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta)$ for some random $\theta \sim p(\theta)$, and

$$\sum_{i=1}^n Y_i \mid \theta \sim \text{Binomial}(n, \theta).$$

Significance

This is the theorem that formally justifies "acting as if" the data are iid given a parameter θ , whenever all one is willing to assume is exchangeability.

Exercise

State the theorem's hypotheses precisely (*infinite* exchangeability, not merely finite- n exchangeability), and note why finite exchangeability alone is not sufficient for the representation to hold exactly.

2.5 The Beta–Bernoulli Model

Model

$$Y_i \mid \theta \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta), \quad y = (y_1, \dots, y_n).$$

$$\underbrace{p(\theta)}_{\text{prior}}, \quad \underbrace{p(y \mid \theta)}_{\text{likelihood}}, \quad \underbrace{p(\theta \mid y)}_{\text{posterior}} \propto p(\theta) p(y \mid \theta).$$

The **Beta density**: for $X \sim \text{Beta}(a, b)$,

$$f_X(x) \propto x^{a-1}(1-x)^{b-1}, \quad 0 < x < 1.$$

Posterior derivation

With likelihood $p(y \mid \theta) = \theta^{\sum y_i} (1 - \theta)^{n - \sum y_i}$, and writing $z = \sum_i y_i$,

$$p(\theta \mid y) \propto p(\theta) \theta^z (1 - \theta)^{n-z}.$$

2.5.1 The Uniform Prior Special Case

Take $\theta \sim U(0, 1)$, i.e. $p(\theta) = 1$ on $(0, 1)$. Then

$$p(\theta \mid y) \propto \theta^z (1 - \theta)^{n-z} \implies \theta \mid y \sim \text{Beta}(z + 1, n - z + 1).$$

Marginal distribution of z under the uniform prior

With $p(\theta) = 1$, the marginal (prior predictive) distribution of $z = \sum_i Y_i$ is **discrete uniform** on $\{0, 1, \dots, n\}$:

$$z \in \{0, 1, \dots, n\}, \quad \text{each with probability } \frac{1}{n+1}.$$

Exercise

Derive this marginal result directly:

$$p(z) = \binom{n}{z} \int_0^1 \theta^z (1 - \theta)^{n-z} d\theta,$$

evaluate the integral using the Beta function $B(z+1, n-z+1) = \frac{z!(n-z)!}{(n+1)!}$, and simplify to confirm $p(z) = \frac{1}{n+1}$ for every $z \in \{0, \dots, n\}$. This is a classical (and slightly surprising) special property of the uniform prior on a Bernoulli success probability.

2.5.2 Posterior Prediction

Goal

Predict a new observation $\tilde{y}$ given the data y .

$$\Pr(\tilde{y} = 1 | y) = \int_0^1 \Pr(\tilde{y} = 1 | \theta, y) p(\theta | y) d\theta = \int_0^1 \theta p(\theta | y) d\theta = E[\theta | y].$$

Using $\theta | y \sim \text{Beta}(z + 1, n - z + 1)$ and the Beta mean formula $E(X) = \frac{a}{a+b}$ for $X \sim \text{Beta}(a, b)$,

$$\Pr(\tilde{y} = 1 | y) = E[\theta | y] = \frac{z + 1}{(z + 1) + (n - z + 1)} = \frac{z + 1}{n + 2}.$$

Laplace's Rule of Succession

$\frac{z + 1}{n + 2}$ is the classical **rule of succession**: compare with the maximum likelihood estimate $\hat{\theta}_{\text{MLE}} = z/n$. The posterior mean shrinks the raw proportion slightly toward $\frac{1}{2}$, with the shrinkage effect vanishing as $n \rightarrow \infty$.

Exercise

Verify z/n vs. $(z + 1)/(n + 2)$ numerically for a few small (z, n) pairs and confirm they converge as n grows. Also derive $\Pr(\tilde{y} = 1 | y) = E[\theta | y]$ in general (not just in this conjugate case) by justifying the interchange of integration order in $\int p(\tilde{y} | \theta) p(\theta | y) d\theta$.

2.5.3 Summarizing the Posterior

For $\theta | y \sim \text{Beta}(z + 1, n - z + 1)$, the standard posterior summaries are:

Quantity	Notation / value
Posterior mean	$E(\theta y) = \frac{z + 1}{n + 2}$
Posterior median	$\text{Med}(\theta y)$
Posterior mode	$\text{Mode}(\theta y)$
Posterior standard deviation	$\text{SD}(\theta y)$
Maximum likelihood estimate	$\hat{\theta}_{\text{MLE}} = z/n$

The posterior density $\text{Beta}(z + 1, n - z + 1)$ is unimodal on $(0, 1)$, rising from 0 at $\theta = 0$ to a peak and back down to 0 at $\theta = 1$:

$$\theta p(\theta | y)^4 * ((1 - x)^2)/0.0119;$$

Figure 2.1: Sketch of a unimodal $\text{Beta}(z + 1, n - z + 1)$ posterior density, skewed according to z relative to n (here $z = 8$, $n = 10$).

Credible interval convention

The values 0.05 and 0.95 (or 0.025 and 0.975) are the standard lower/upper quantiles used to construct a two-sided 90% (resp. 95%) posterior **credible interval** for θ : $(\theta_{0.05}, \theta_{0.95})$ or $(\theta_{0.025}, \theta_{0.975})$, depending on the convention in use.

Exercise

- Derive the mode of $\text{Beta}(a, b)$ for $a, b > 1$: $\text{Mode} = \frac{a-1}{a+b-2}$, and specialize to $a = z+1$, $b = n-z+1$.
- Derive the variance of $\text{Beta}(a, b)$: $\text{Var}(X) = \frac{ab}{(a+b)^2(a+b+1)}$, specialize similarly, and take the square root for $\text{SD}(\theta | y)$.
- Compare $\hat{\theta}_{\text{MLE}} = z/n$, the posterior mean $(z+1)/(n+2)$, and the posterior mode z/n : note that under a *uniform* prior the posterior mode coincides with the MLE, while the posterior mean does not — confirm this coincidence algebraically.
- Numerically integrate (or use the incomplete Beta function) to compute the 0.05 and 0.95 quantiles of $\text{Beta}(z+1, n-z+1)$ for a specific (z, n) from class, to build intuition for how the credible interval narrows as n grows.

Chapter 3

Week 3: Conjugate Priors — Beta-Bernoulli (General Case) and Normal-Normal

Lecture 5

3.0.1 The Beta-Bernoulli Conjugate Model

Let

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Ber}(\theta), \quad \theta \sim U(0, 1).$$

Then

$$p(\theta \mid x_1, \dots, x_n) \sim \text{Beta}(\sum X_i + 1, n - \sum X_i + 1).$$

$$p(\theta \mid x_1 \dots x_n) \propto \theta^{\sum X_i} (1 - \theta)^{n - \sum X_i}$$

$$\text{Beta}(a, b) \propto \theta^{a-1} (1 - \theta)^{b-1}$$

Definition 3.1 (Conjugate prior). If the prior distribution is of the same family as the posterior distribution, the prior is called a **conjugate prior**.

For a general $\text{Beta}(\alpha, \beta)$ prior,

$$p(\theta \mid X_1 \dots X_n) \propto p(\theta) p(X_1 \dots X_n \mid \theta) \propto \theta^{\alpha + \sum X_i - 1} (1 - \theta)^{\beta + (n - \sum X_i) - 1},$$

$$\therefore \theta \mid X_1, \dots, X_n \sim \text{Beta}\left(\sum X_i + \alpha, n - \sum X_i + \beta\right).$$

The posterior predictive probability of a new success is

$$p(\tilde{y} = 1 \mid X_1 \dots X_n) = E(\theta \mid X_1 \dots X_n) = \frac{\alpha + \sum X_i}{n + \alpha + \beta} \xrightarrow{n \rightarrow \infty} \frac{\sum X_i}{n}.$$

General setup

$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Ber}(\theta)$. If $p(\theta) \propto \theta^{\alpha-1} (1 - \theta)^{\beta-1}$ [prior], then $\theta \sim \text{Beta}(\alpha, \beta)$.

$$\hat{\theta}_{\text{MLE}} = \frac{\sum X_i}{n}, \quad \text{Var}(\theta \mid X_1 \dots X_n) = \frac{(\alpha + \sum X_i)(\beta + n - \sum X_i)}{(\alpha + \beta + n)^2 (\alpha + \beta + n + 1)}.$$

As $n \rightarrow \infty$,

$$\text{Var}(\theta \mid X_1 \dots X_n) \approx \frac{(\sum X_i)(n - \sum X_i)}{n^3} \rightarrow 0,$$

i.e. the posterior concentrates around the truth as the sample size grows, regardless of the prior chosen.

3.0.2 Normal Distribution with Known Variance

$$p(y \mid \theta) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2\sigma^2}(y - \theta)^2\right).$$

Taking a normal prior $p(\theta) = N(\mu_0, \tau_0^2)$,

$$p(\theta \mid y) \propto \underline{p(\theta)} \exp\left(-\frac{1}{2\sigma^2}(y - \theta)^2\right) \propto \exp\left(-\frac{1}{2\tau_0^2}(\theta - \mu_0)^2\right) \exp\left(-\frac{1}{2\sigma^2}(y - \theta)^2\right).$$

Completing the square in θ (i.e. collecting terms of the form $\exp(A\theta^2 + B\theta + C)$) shows that the posterior is again normal:

$$\begin{aligned} \theta \mid y &\sim N(\mu_1, \tau_1^2), \quad \text{where} \\ \mu_1 &= \frac{\frac{1}{\tau_0^2}\mu_0 + \frac{1}{\sigma^2}y}{\frac{1}{\tau_0^2} + \frac{1}{\sigma^2}} \quad \text{and} \quad \frac{1}{\tau_1^2} = \frac{1}{\tau_0^2} + \frac{1}{\sigma^2}. \end{aligned}$$

The n -observation case

Now let $y_1, \dots, y_n \stackrel{\text{iid}}{\sim} N(\theta, \sigma^2)$. Since $\bar{y} \sim N(\theta, \sigma^2/n)$ is sufficient for θ ,

$$p(\theta \mid y_1, \dots, y_n) \equiv p(\theta \mid \bar{y}) \propto p(\theta) p(\bar{y} \mid \theta),$$

$$\theta \mid \bar{y} \sim N(\mu_1, \tau_1^2), \quad \text{where} \quad \mu_1 = \frac{\frac{\mu_0}{\tau_0^2} + \frac{n}{\sigma^2}\bar{y}}{\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}}, \quad \frac{1}{\tau_1^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}.$$

Here $\bar{y}$ is an estimator of θ , and equivalently

$$p(y \mid \theta) \propto \exp\left\{-\frac{\sum (y_i - \theta)^2}{2\sigma^2}\right\} \propto \exp\left\{-\frac{(\bar{y} - \theta)^2}{2\sigma^2/n}\right\}.$$

Special case: n

$$\mu_1 = \frac{\frac{\mu_0}{\tau_0^2} + \frac{y}{\sigma^2}}{\frac{1}{\tau_0^2} + \frac{1}{\sigma^2}}, \quad \bar{y} = y, \quad \text{Precision} = \frac{1}{\text{Variance}}.$$

- $\mu_1 \equiv \mu_0$ if $y = \mu_0$, or if $\tau_0^2 = 0$ (and $\sigma^2 \neq 0$).
- $\mu_1 = y$ if $y = \mu_0$, or if $\sigma^2 = 0$ (and $\tau_0^2 \neq 0$).

- μ_1 is always a **weighted average** of the prior mean μ_0 and the data y , weighted by their respective precisions.
- If $\mu_0 = 0$, then $\mu_1 = \frac{y \tau_0^2}{\sigma^2 + \tau_0^2}$, and with n observations, $\mu_1 = \frac{y \tau_0^2 / n}{\sigma^2 / n + \tau_0^2}$, using $\tau_0^2(n) = \sigma^2 / n$.

$$p(\theta | y) \propto p(\theta) \prod_{i=1}^n p(y_i | \theta)$$

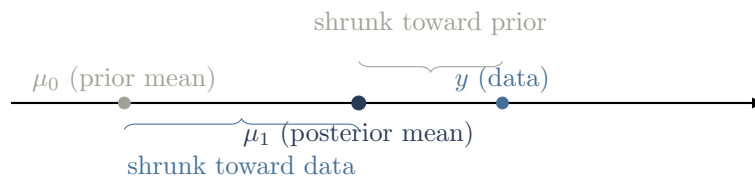


Figure 3.1: The posterior mean μ_1 is a precision-weighted average, "shrunk" from the prior mean μ_0 toward the observed data y (and vice versa). As $\sigma^2 \rightarrow 0$ (perfectly precise data), $\mu_1 \rightarrow y$; as $\tau_0^2 \rightarrow 0$ (a dogmatic prior), $\mu_1 \rightarrow \mu_0$.

3.0.3 Predictive Distribution

$$p(\tilde{y} | y) = \int p(\tilde{y} | \theta) p(\theta | y) d\theta \propto \int \exp\left(-\frac{1}{2\sigma^2}(\tilde{y} - \theta)^2\right) \exp\left(-\frac{1}{2\tau_1^2}(\theta - \mu_1)^2\right) d\theta,$$

which is recognized as (proportional to) the pdf of a bivariate normal random vector $(\tilde{y}, \theta)$, integrated over θ .

Question posed in lecture

If $\theta \sim N(\mu_0, \tau_0^2)$ and $y | \theta \sim N(\theta, \sigma^2)$, what is the **marginal** distribution $p(y)$? This is answered at the start of Lecture 6 below.

Lecture 6

3.0.4 Recap and the Marginal Distribution of y

From Lecture 5,

$$\theta | x_1, \dots, x_n \sim N(\mu_1, \tau_1^2), \quad \mu_1 = \frac{\frac{1}{\tau_0^2} \mu_0 + \frac{n}{\sigma^2} \bar{y}}{\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}}.$$

The joint density of (θ, y) is (jointly) Gaussian:

$$p(y, \theta) \propto \exp\left\{-\frac{1}{2} \begin{pmatrix} \theta \\ y \end{pmatrix}^T \Sigma^{-1} \begin{pmatrix} \theta \\ y \end{pmatrix}\right\} \quad (\text{up to a mean shift by } \mu).$$

Setup

$$\theta \sim N(\mu_0, \tau_0^2), \quad y | \theta \sim N(\theta, \sigma^2). \quad \text{Find } p(y).$$

$$p(y) = \int p(y | \theta) p(\theta) d\theta = \int \frac{\exp\left(-\frac{1}{2\sigma^2}(y - \theta)^2\right)}{\sigma\sqrt{2\pi}} \cdot \frac{\exp\left(-\frac{1}{2\tau_0^2}(\theta - \mu_0)^2\right)}{\tau_0\sqrt{2\pi}} d\theta.$$

Writing $y | \theta = \theta + N(0, \sigma^2)$, i.e. $y = \theta + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2)$ independent of $\theta \sim N(\mu_0, \tau_0^2)$, the marginal of a sum of two independent normals is again normal, with means and variances adding:

$$y \sim N(\mu_0, \tau_0^2 + \sigma^2).$$

Why

$\begin{pmatrix} \theta \\ \varepsilon \end{pmatrix}$ is a bivariate normal random vector with independent components, so any linear combination — here $y = \theta + \varepsilon$ — is itself normal, with mean $\mu_0 + 0 = \mu_0$ and variance $\tau_0^2 + \sigma^2$.

3.0.5 The Joint (Multivariate Normal) Formulation

Let $y_1, \dots, y_n | \theta \stackrel{\text{iid}}{\sim} N(\theta, \sigma^2)$ and $\theta \sim N(\mu_0, \tau_0^2)$. What is $p(y_1, \dots, y_n)$?

Write $y_i = \theta + \varepsilon_i$ for $i = 1, \dots, n$, where $\varepsilon_1, \dots, \varepsilon_n$ are iid $N(0, \sigma^2)$, independent of $\theta \sim N(\mu_0, \tau_0^2)$. Then

$$\begin{pmatrix} \theta \\ \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix} \sim MVN(\mathcal{M}, \Sigma), \quad \mathcal{M} = \begin{pmatrix} \mu_0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \Sigma = \text{diag}(\tau_0^2, \sigma^2, \dots, \sigma^2).$$

Since $y = A \begin{pmatrix} \theta \\ \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix}$ is a linear transformation, with

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 & \cdots \\ 1 & 0 & 1 & 0 & \cdots \\ 1 & 0 & 0 & 1 & \cdots \\ \vdots & & & & \ddots \end{pmatrix},$$

the standard multivariate-normal linear-transformation rule gives

$$y = A \begin{pmatrix} \theta \\ \varepsilon \end{pmatrix} \implies y \sim MVN(A\mathcal{M}, A\Sigma A^T).$$

Since $\text{Cov}(\theta + \varepsilon_i, \theta + \varepsilon_j) = \text{Var}(\theta) = \tau_0^2$ for $i \neq j$ (the ε_i are independent and contribute nothing to the covariance), this works out to

$$\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \sim \text{MVN} \left(\begin{pmatrix} \mu_0 \\ \vdots \\ \mu_0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 + \sigma^2 & \tau_0^2 & \cdots & \tau_0^2 \\ \tau_0^2 & \tau_0^2 + \sigma^2 & & \vdots \\ \vdots & & \ddots & \\ \tau_0^2 & \cdots & & \tau_0^2 + \sigma^2 \end{pmatrix} \right).$$

$\tau_0^2 + \sigma^2$	τ_0^2	τ_0^2	τ_0^2
τ_0^2	$\tau_0^2 + \sigma^2$	τ_0^2	τ_0^2
τ_0^2	τ_0^2	$\tau_0^2 + \sigma^2$	τ_0^2
τ_0^2	τ_0^2	τ_0^2	$\tau_0^2 + \sigma^2$

Figure 3.2: Structure of the induced covariance matrix for $y_1, \dots, y_n \sim \text{MVN}(\mu_0 \mathbf{1}, \Sigma)$: diagonal entries equal $\tau_0^2 + \sigma^2$, off-diagonal entries equal τ_0^2 — every pair of observations shares the common uncertainty about θ .

$$\theta \mid y_1, \dots, y_n \sim N(\mu_1, \tau_1^2), \quad y \mid \theta \sim N(\theta, \sigma^2), \quad y \mid \theta = \underline{Z} + \underline{\theta}, \quad Z \sim N(0, \sigma^2).$$

3.0.6 Prediction for Unobserved Data

Let $\theta \sim N(\mu_0, \tau_0^2)$ and $y_1, \dots, y_M \mid \theta \stackrel{\text{iid}}{\sim} N(\theta, \sigma^2)$. Given $y_1, \dots, y_n$ (observed), with y_M (and more generally $y_{n+1}, \dots, y_M$) unobserved:

$$p(y_M \mid y_1, \dots, y_n) = \int p(y_M \mid \theta) p(\theta \mid y_1, \dots, y_n) d\theta, \quad y_M \mid y_1, \dots, y_n \sim N(\mu_1, \sigma^2 + \tau_1^2).$$

For predicting several future observations at once,

$$y_{n+1}, \dots, y_M \mid y_1, \dots, y_n \longrightarrow \sum_{k=1}^{M-n} y_{n+k} \mid y_1, \dots, y_n \sim N((M-n)\mu_1, (M-n)\sigma^2 + (M-n)^2\tau_1^2).$$

As more data accumulate,

$$\tau_1^2 = \frac{1}{\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}} \longrightarrow 0 \quad (n \rightarrow \infty),$$

so the posterior for θ collapses onto the true value: $p(\theta \mid y_1, \dots, y_n) \rightarrow \delta_{\theta_0}$ as $n \rightarrow \infty$ — the Bayesian analogue of consistency.

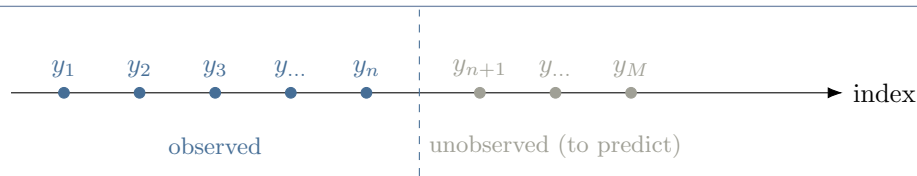


Figure 3.3: Schematic of the prediction task: given observed $y_1, \dots, y_n$, predict the remaining unobserved $y_{n+1}, \dots, y_M$ by integrating over the posterior $p(\theta \mid y_1, \dots, y_n)$.

3.0.7 The Beta-Bernoulli Model Revisited (General α, β)

$$X_1, \dots, X_n \mid \theta \stackrel{\text{iid}}{\sim} \text{Ber}(\theta), \quad \theta \sim \text{Beta}(\alpha, \beta), \quad P(X_1 = 1) = \frac{\alpha}{\alpha + \beta}.$$

The prior predictive (marginal) probability of any specific sequence is

$$P(X_1, \dots, X_n) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 \theta^{\sum X_i + \alpha - 1} (1 - \theta)^{n - \sum X_i + \beta - 1} d\theta = \frac{B(\alpha + \sum X_i, \beta + n - \sum X_i)}{B(\alpha, \beta)}.$$

3.0.8 A Note on Independence

In general,

$$P(X_1 = 1, X_2 = 1) \neq P(X_1 = 1) P(X_2 = 1),$$

because the X_i are only *conditionally* independent given θ ; marginally (after integrating out θ), they are dependent — this is the same phenomenon as Section 2.3 above. A small table illustrating dependent binary co-occurrence patterns:

1	0	0	0
1	1	0	0
0	0	1	1
0	1	0	1

Sampling algorithms mentioned in passing

Metropolis-Hastings algorithm • MCMC (Markov Chain Monte Carlo) • HMC (Hamiltonian Monte Carlo) — computational tools for posterior sampling when no closed-form conjugate posterior is available; covered in detail later in the course.

3.0.9 Exponential Family of Distributions

Definition

A distribution belongs to the (one-parameter) **exponential family** if it can be written

$$p(y_i \mid \theta) = f(y_i) g(\theta) \exp \{ \phi(\theta)^T u(y_i) \},$$

e.g. Poisson, Normal, Binomial, Exponential.

For an iid sample,

$$p(y_1, \dots, y_n \mid \theta) = \prod_{i=1}^n f(y_i) (g(\theta))^n \exp \left\{ \phi(\theta)^T \sum_{i=1}^n u(y_i) \right\}.$$

If the prior is chosen in the matching conjugate family,

$$p(\theta) \propto (g(\theta))^\eta \exp \{ \phi(\theta)^T \nu \},$$

then the posterior stays within the same family:

$$p(\theta \mid y) \propto p(\theta) \left[(g(\theta))^n \exp \{ \phi(\theta)^T \sum_i u(y_i) \} \right] \propto (g(\theta))^{\eta+n} \exp \left\{ \phi(\theta)^T \left(\sum_i u(y_i) + \nu \right) \right\}.$$

Exercise

Show that the Binomial distribution is a member of the exponential family. Hence identify an appropriate conjugate prior for it. (Hint: the natural parameter is $\log(\theta/(1-\theta))$, and the conjugate prior is $\text{Beta}(\alpha, \beta)$, giving $E[\theta] = \alpha/(\alpha + \beta)$.)

Chapter 4

Week 4: Improper Priors, Jeffreys Prior, and Kullback–Leibler Divergence

Lecture 7 — August 11

4.0.1 Conjugate Prior: Normal–Normal (Mean Unknown, Variance Known)

$\theta \sim N(\mu_0, \tau_0^2)$. The prior should have a suitably **high variance** τ_0^2 when little is known about θ a priori.

Motivating range

E.g. modeling human heights in the range 100–200 cm: $Y_1, \dots, Y_n \mid \theta \sim N(\theta, \sigma^2)$.

$$\pi(\theta \mid Y) \propto \frac{1}{\sqrt{2\pi\tau_0^2}} \exp\left\{-\frac{(\theta - \mu_0)^2}{2\tau_0^2}\right\} \cdot \frac{1}{(\sigma\sqrt{2\pi})^n} \exp\left\{-\frac{\sum(Y_i - \theta)^2}{2\sigma^2}\right\}.$$

As the prior becomes increasingly flat, $\tau_0 \rightarrow \infty$,

$$\pi(\theta \mid Y_1, \dots, Y_n) \approx \frac{1}{(\sigma\sqrt{2\pi})^n} \exp\left\{-\frac{(\theta - \bar{Y})^2}{2\sigma^2/n}\right\},$$

i.e. the posterior is driven almost entirely by the likelihood, recovering $\theta \mid Y \approx N(\bar{Y}, \sigma^2/n)$ — the Bayesian analogue of the classical sampling distribution of $\bar{Y}$. In general,

$$p(Y_1, \dots, Y_n) = p(Y_1, \dots, Y_n \mid \theta) \pi(\theta).$$

Example priors and regression extensions

Example priors: $N(0, 100)$ (vague/diffuse) vs. $N(0, 10)$ (more informative).
The same conjugate-updating logic extends to regression settings, e.g.

$$y = \beta_0 + \beta_1 x + \varepsilon, \quad y = \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon,$$

where each regression coefficient can be assigned its own (conditionally conjugate) normal prior.

4.0.2 Does the Prior Need to Be a Valid Probability Density?

Answer

Prior: No. **Posterior:** Yes (it must integrate to 1 to be a valid probability distribution).

$$\pi(\theta | y) \propto \pi(\theta) \pi(y | \theta), \quad \pi(\theta | x) = \frac{\pi(\theta) \pi(x | \theta)}{\int \pi(\theta) \pi(x | \theta) d\theta}.$$

The unnormalized statement $\pi(\theta | A) \propto \pi(\theta) \pi(A | \theta)$ need not integrate to 1 on its own. But

$$\int \pi(\theta | x) d\theta = 1 \quad \text{provided} \quad \int \pi(\theta) \pi(x | \theta) d\theta < \infty.$$

4.0.3 Improper Priors

Definition 4.1 (Improper prior). A prior $\pi(\theta)$ is **improper** if $\int_{-\infty}^{\infty} \pi(\theta) d\theta = \infty$ (so it cannot be normalized to a proper density), yet it may still be used formally provided the resulting posterior integral is finite.

Writing $C = \int \pi(\theta) d\theta \leq \infty$: if $C < \infty$ the prior is proper (after normalizing by $1/C$); an improper prior corresponds to $C = \infty$, but the posterior

$$\frac{1}{C} \int \pi(\theta) \pi(x | \theta) d\theta$$

can still be a legitimate (proper) posterior distribution as long as $\int \pi(\theta) \pi(x | \theta) d\theta < \infty$, i.e. $\int \pi(\theta, x) d\theta$ is finite even though $\int \pi(\theta) d\theta$ is not.

Example 1: flat prior on a normal mean

$$X | \theta \sim N(\theta, 1), \quad \pi(\theta) \propto 1 \quad \Rightarrow \quad \int \pi(\theta) d\theta = \infty \quad (\text{improper prior}).$$

But after observing X ,

$$\pi(\theta | x) \propto \pi(\theta) f(x | \theta) \propto \exp\left\{-\frac{(\theta - x)^2}{2}\right\} \quad \Rightarrow \quad \theta | x \sim N(x, 1),$$

which *is* a valid (proper) posterior — illustrating that an improper prior can still yield a perfectly sensible, proper posterior.

Example 2: flat prior on the upper endpoint of a Uniform

$$X | \theta \sim U(0, \theta), \quad \pi(\theta) \propto 1, \quad \pi(x | \theta) = \frac{1}{\theta} \mathbf{1}\{\theta > x\}.$$

Then

$$\pi(\theta | x) \propto \pi(\theta) \pi(x | \theta) \propto \frac{1}{\theta} \mathbf{1}\{\theta > x\}.$$

- If instead $\pi(\theta) \propto \frac{1}{\theta}$, one obtains a **proper posterior**.
- With the flat prior $\pi(\theta) \propto 1$, one instead obtains an **improper posterior** (the relevant

integral $\int \pi(\theta) \pi(x | \theta) d\theta$ diverges) — so *not every* improper prior leads to a valid posterior; each case must be checked.

A caution about diffuse — but proper — priors

Even a *proper* prior with very large variance can behave in ways that seem counter-intuitive. E.g. for $\theta \sim N(\mu_0, \sigma_0^2)$, comparing the prior density at $\theta = \mu_0$ and at $\theta = \mu_0 + 10$:

$$\pi(\mu_0) = \frac{1}{\sigma_0 \sqrt{2\pi}}, \quad \pi(\mu_0 + 10) = \frac{1}{\sigma_0 \sqrt{2\pi}} \exp\left\{-\frac{10^2}{2\sigma_0^2}\right\},$$

$$\frac{\pi(\mu_0)}{\pi(\mu_0 + 10)} = \exp\left(\frac{10^2}{2\sigma_0^2}\right),$$

which is enormous for small σ_0^2 but close to 1 for very large σ_0^2 — showing how the choice between, say, $N(0, 10)$ and $N(0, 100)$ can materially affect how "non-informative" the prior actually is in relative terms, even though $\int \pi(\theta) d\theta \neq 1$ is not literally the issue here (both are proper).

4.0.4 Conjugate Priors for Other Exponential Families

$Y_1, \dots, Y_n | \sigma^2 \sim N(\mu, \sigma^2)$, μ known. What is $p(\sigma^2)$ (conjugate prior)?

Binomial: derive its conjugate prior by expressing it in exponential-family form.

Poisson: derive its conjugate prior by expressing it in exponential-family form.

Hints

- Identify the natural parameter and the sufficient statistic.
- Construct a prior over these natural parameters.
- Use a Jacobian to convert that prior into one for the usual parameter of the distribution.
- (Hint: think along the lines of the Beta prior for the Bernoulli parameter.)
- General form: $\pi(\theta) \propto \exp\{\eta(\theta)^T T(x) - A(\theta)\}$; derive $\pi(\lambda)$ for the Poisson using its exponential-family representation.

Lecture 8 — August 12, 2026

4.0.5 Poisson Regression with a Covariate

Setup

Given paired data $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, model

$$y_i | x_i \sim \text{Poisson}(\theta x_i).$$

The likelihood is

$$L(\theta) = \prod_{i=1}^n \frac{e^{-\theta x_i} (\theta x_i)^{y_i}}{y_i!} \propto e^{-\theta \sum x_i} \theta^{\sum y_i}.$$

Roadmap for a full Bayesian analysis

- 1) Write down the likelihood.
- 2) Choose a conjugate prior (or another appropriate form).
- 3) Derive the posterior.
- 4) Derive the predictive distribution.

4.0.6 Non-informative Priors and Reparametrization

Setup

$X | \theta \sim N(\theta, \sigma^2)$, with a flat prior $\pi(\theta) \propto 1$. Under a one-to-one reparametrization $\theta = e^\gamma$, is $\pi(\gamma) \propto 1$ also "non-informative"?

$$\frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$$

Non-informative prior under reparametrization. If $\phi = g(\theta)$ is a one-to-one transformation, then the induced density is

$$\pi(\phi) = \pi(g^{-1}(\phi)) |J|,$$

where J is the Jacobian of the inverse transformation.

The problem this raises

A flat prior is *not* invariant under reparametrization: $\pi(\theta) \propto 1$ does *not*, in general, imply $\pi(\phi) \propto 1$ for $\phi = g(\theta)$, because of the Jacobian factor. This motivates a principled way of constructing "objective" priors that *are* invariant to the choice of parametrization — Jeffreys' prior.

4.0.7 Jeffreys Prior

Guiding principle

Any rule for determining the prior density should yield an *equivalent* result if applied to a transformed parameter: i.e. $p(\phi)$, computed by first determining $p(\theta)$ and then transforming, should match the distribution obtained by applying the same rule directly to ϕ .

Let $\phi = h(\theta)$ for a one-to-one h . Then

$$p(\phi) = p(\theta) \left| \frac{d\theta}{d\phi} \right| = p(\theta) |h'(\theta)|^{-1}.$$

Jeffreys' principle:

$$\pi(\theta) \propto \sqrt{I(\theta)}, \quad I(\theta) = -E \left(\frac{\partial^2 \log f(y | \theta)}{\partial \theta^2} \right) \quad (\text{Fisher information}).$$

Under this rule, applying the transformation consistently gives

$$\pi(\theta) \propto \sqrt{I(\theta)} = \sqrt{I(\phi)} \left| \frac{d\phi}{d\theta} \right|, \quad I(\phi) = -E \left(\frac{\partial^2 \log f(y | \phi)}{\partial \phi^2} \right),$$

and correspondingly

$$E \left[\left(\frac{\partial \log f(y | \theta)}{\partial \theta} \right)^2 \right] = I(\theta) \left| \frac{d\theta}{d\phi} \right|^2,$$

so that the Jeffreys rule applied to θ and then transformed to ϕ agrees with the Jeffreys rule applied directly to ϕ — the defining invariance property.

Binomial example

For $X | \theta \sim \text{Bin}(n, p)$, taking $\theta = \log(p/(1-p))$ (the log odds) and a flat prior $\pi(\theta) \propto 1$ on the log-odds scale corresponds to a very different (non-flat) prior on p . Directly, the Fisher information for p is

$$-E \left(\frac{\partial^2 \log f(y | p)}{\partial p^2} \right) = \frac{1}{p(1-p)},$$

so the Jeffreys prior for p is $\pi(p) \propto \sqrt{\frac{1}{p(1-p)}} = \text{Beta}(\frac{1}{2}, \frac{1}{2})$ (up to normalization) — automatically invariant to which parametrization (p , $\theta = \log(p/(1-p))$, etc.) one starts from.

4.0.8 Kullback–Leibler Divergence and Its Connection to Jeffreys’ Prior

Definition 4.2 (Kullback–Leibler divergence).

$$D_{\text{KL}}(P \parallel Q) = E_P \left[\log \frac{p}{q} \right].$$

Local expansion linking D_{KL} to Fisher information

Let $L(\theta) = \log f(y | \theta)$, and consider a small perturbation $\theta \rightarrow \theta + d\theta$. A Taylor expansion gives

$$L(\theta + d\theta) = L(\theta) + L'(\theta) d\theta + \frac{1}{2} L''(\theta) (d\theta)^2 + o((d\theta)^2).$$

Taking expectations under the true θ : the first-order term vanishes on average, $-E_\theta[L'(\theta)] d\theta$ contributes at leading order, and the curvature term contributes

$$-\frac{1}{2} E_\theta[L''(\theta)] (d\theta)^2 + o((d\theta)^2) = \frac{1}{2} I(\theta) (d\theta)^2 + o((d\theta)^2),$$

using $I(\theta) = -E_\theta[L''(\theta)]$. Combining these,

$$D_{\text{KL}}(P(X | \theta), P(X | \theta + d\theta)) = 0 + I(\theta) (d\theta)^2 + o((d\theta)^2).$$

Interpretation

The Fisher information $I(\theta)$ measures how quickly the model distribution changes (in KL divergence) as θ moves an infinitesimal distance $d\theta$ — i.e. it is a local notion of “distinguishability” between nearby parameter values, given a single observation $y \sim f(y | \theta)$. Jeffreys’ prior $\pi(\theta) \propto \sqrt{I(\theta)}$ can therefore be read as placing prior mass *uniformly with respect to this distinguishability metric* rather than uniformly on the raw parameter scale — which is exactly why it is invariant to reparametrization.

$$\pi(\theta) \propto \sqrt{I(\theta)}$$

4.0.9 Worked Example: The Poisson Model

For $Y \mid \lambda \sim \text{Poisson}(\lambda)$, i.e. $p(y \mid \lambda) = \frac{e^{-\lambda} \lambda^y}{y!}$, the Fisher information is

$$I(\lambda) = \frac{1}{\lambda}, \quad \sqrt{I(\lambda)} = \frac{1}{\sqrt{\lambda}}.$$

The Jeffreys prior for the Poisson rate is therefore

$$\pi(\lambda) \propto \frac{1}{\sqrt{\lambda}},$$

i.e. (up to normalization) a $\text{Gamma}(\frac{1}{2}, 0)$ -type improper prior. Equivalently, on the square-root scale $\sqrt{\lambda}$, this prior is flat.

Numerical illustration

Because $\pi(\lambda) \propto \lambda^{-1/2}$ is flat on the $\sqrt{\lambda}$ scale, equal prior mass is assigned to *equal-length intervals in $\sqrt{\lambda}$* , not to equal-length intervals in λ . For instance, the interval $\lambda \in (0, 2.25)$ corresponds to $\sqrt{\lambda} \in (0, 1.5)$, and $\lambda \in (2.25, 6.25)$ corresponds to $\sqrt{\lambda} \in (1.5, 2.5)$ — two intervals of *equal* length (1 unit) on the $\sqrt{\lambda}$ scale, and hence of *equal* Jeffreys prior mass, despite having very different lengths (2.25 vs. 4.00) on the original λ scale.

Appendix: Quick-Reference Formula Sheet

Bayes' theorem (general form).

$$p(\theta | y) = \frac{p(\theta) p(y | \theta)}{p(y)}, \quad p(y) = \int p(\theta) p(y | \theta) d\theta, \quad p(\theta | y) \propto p(\theta) p(y | \theta).$$

Beta–Bernoulli model. $X_1, \dots, X_n | \theta \stackrel{\text{iid}}{\sim} \text{Ber}(\theta)$, $\theta \sim \text{Beta}(\alpha, \beta)$:

$$\theta | X_1, \dots, X_n \sim \text{Beta}(\alpha + \sum X_i, \beta + n - \sum X_i), \quad E[\theta | X] = \frac{\alpha + \sum X_i}{n + \alpha + \beta}.$$

Normal–Normal model. $y_1, \dots, y_n | \theta \stackrel{\text{iid}}{\sim} N(\theta, \sigma^2)$, $\theta \sim N(\mu_0, \tau_0^2)$:

$$\theta | y \sim N(\mu_1, \tau_1^2), \quad \mu_1 = \frac{\frac{\mu_0}{\tau_0^2} + \frac{n}{\sigma^2} \bar{y}}{\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}}, \quad \frac{1}{\tau_1^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}.$$

Jeffreys prior.

$$\pi(\theta) \propto \sqrt{I(\theta)}, \quad I(\theta) = -E\left(\frac{\partial^2 \log f(y | \theta)}{\partial \theta^2}\right), \quad \text{invariant to reparametrization.}$$

Exponential family & conjugacy.

$$p(y | \theta) = f(y) g(\theta) \exp\{\phi(\theta)^T u(y)\} \implies p(\theta) \propto (g(\theta))^\eta \exp\{\phi(\theta)^T \nu\} \implies p(\theta | y) \propto (g(\theta))^{\eta+n} \exp\{\phi(\theta)^T (\nu + u(y))\}$$